

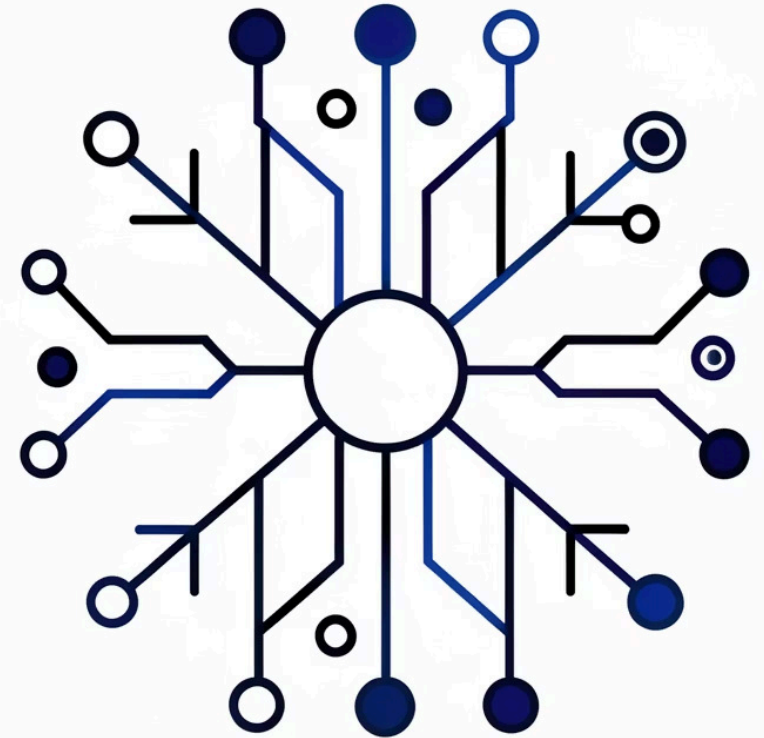
Salary Prediction

A Deep Learning and LLM Agents Approach

Igor Scholz, Antoni Sommer, Olaf Serafin

Silesian University of Technology, Faculty of Applied Mathematics

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CHAPTER 1

Problem Definition



Main Problem

Many job postings do not disclose salary ranges, making it difficult for candidates to evaluate offers.



Target Audience

Job Seekers: Helps estimate expected salary when data is missing.

Recruiters: Supports in defining competitive offers.



Our Solution

An AI system combining neural networks and AI agents to predict annual salaries in USD.

Dataset Overview

Source and Rationale

Source: Glassdoor Jobs dataset (Data Science job postings).

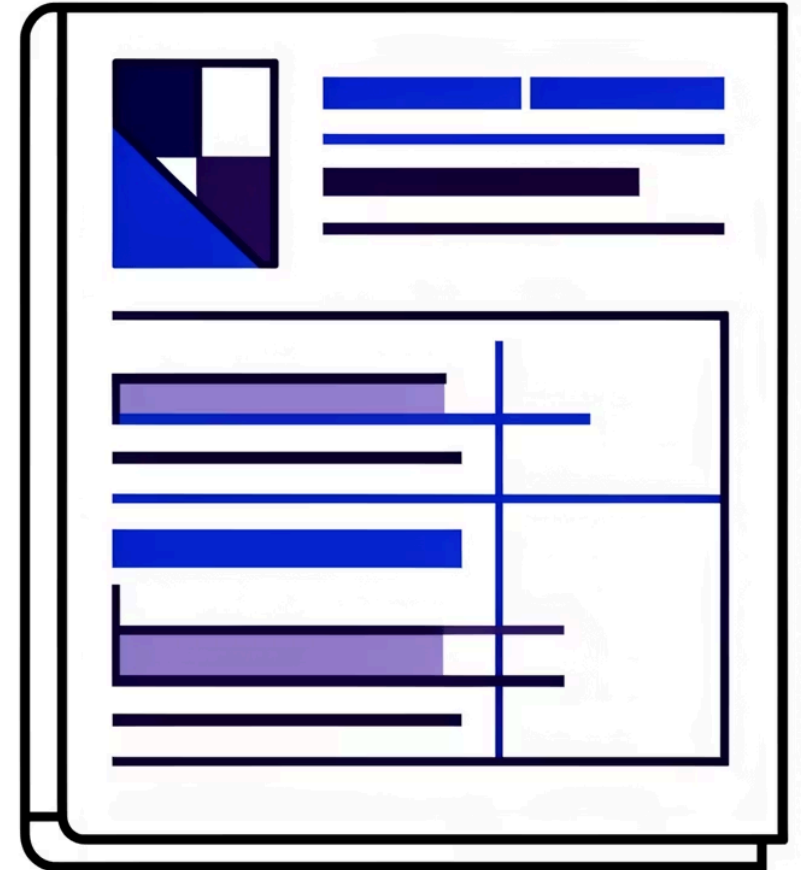
Why this dataset? We chose it because it was the **largest available** and offered the clearest and most structured data compared to alternatives.

Limitations

Despite being the best choice, the dataset size is likely **still too small** for deep learning to reach its full potential, leading to variance challenges.

Key Challenge

- ❏ **Feature Reduction:** The main technical challenge was **feature selection and reduction**. We had to balance model accuracy with the AI agent's ability to reliably extract data without hallucinations.



Final Feature Set

Features optimized for agent reliability:



Position Information

- Job Title
- Job Description
- State (Location)



Company Data

- Company Size
- Sector
- Competitors



Technical Skills

- Python
- Spark
- AWS
- Excel

Value Proposition



Beyond Simple Regression

Unlike standard statistical models, our system actively **enriches data** using external knowledge.



Extraction

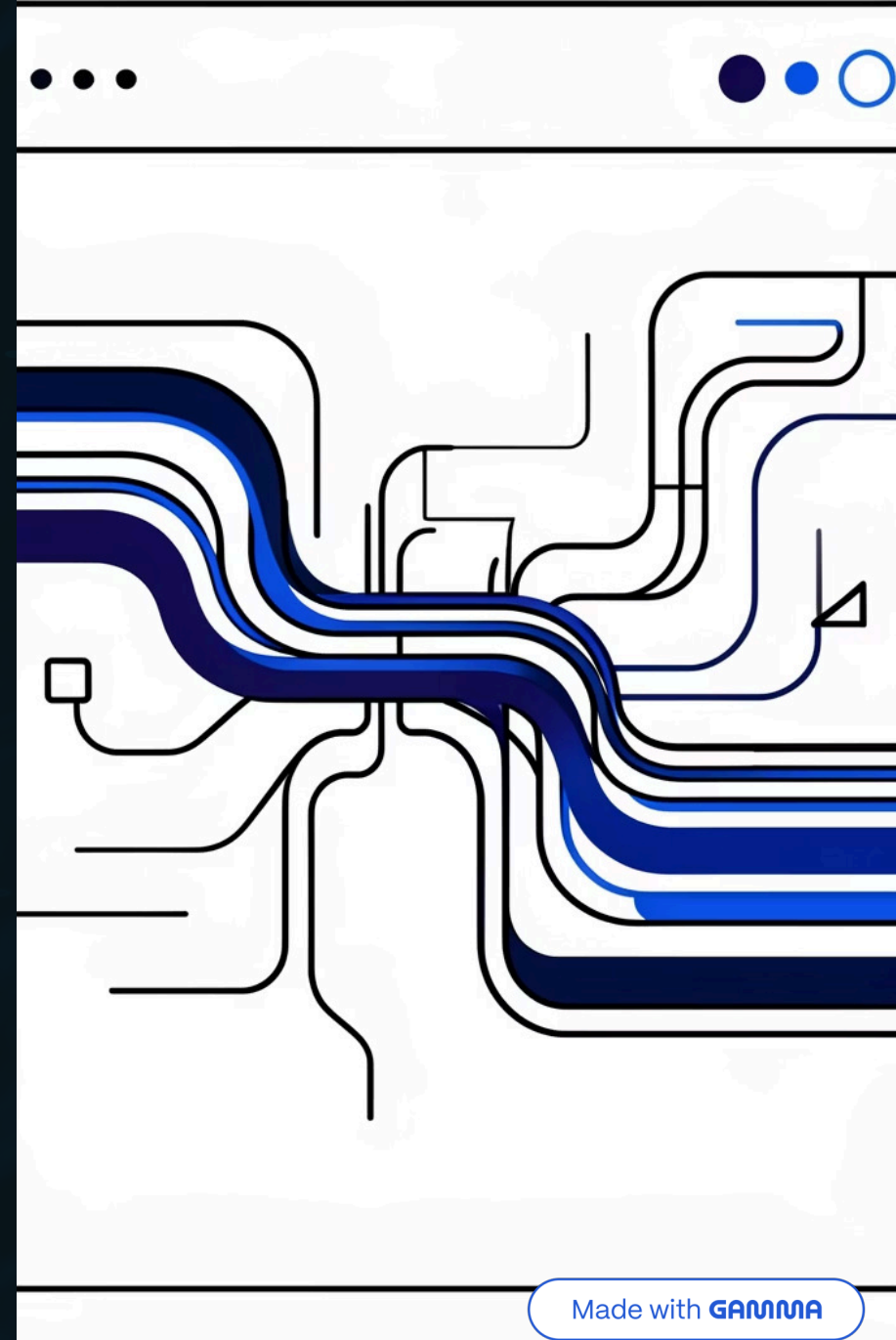
Automatically identifies key details from unstructured job descriptions.



Enrichment

Fills in missing company data (e.g., sector, size) using AI agents **leveraging Google Search**.

Key Benefit: Equips users with a data-driven benchmark for salary negotiation and budgeting.



System Architecture

Workflow in the pipeline:

01

User Interface

The user provides the job offer description through a dedicated frontend.

02

AI Agents Layer

Processes text, retrieves external data, and structures the input.

03

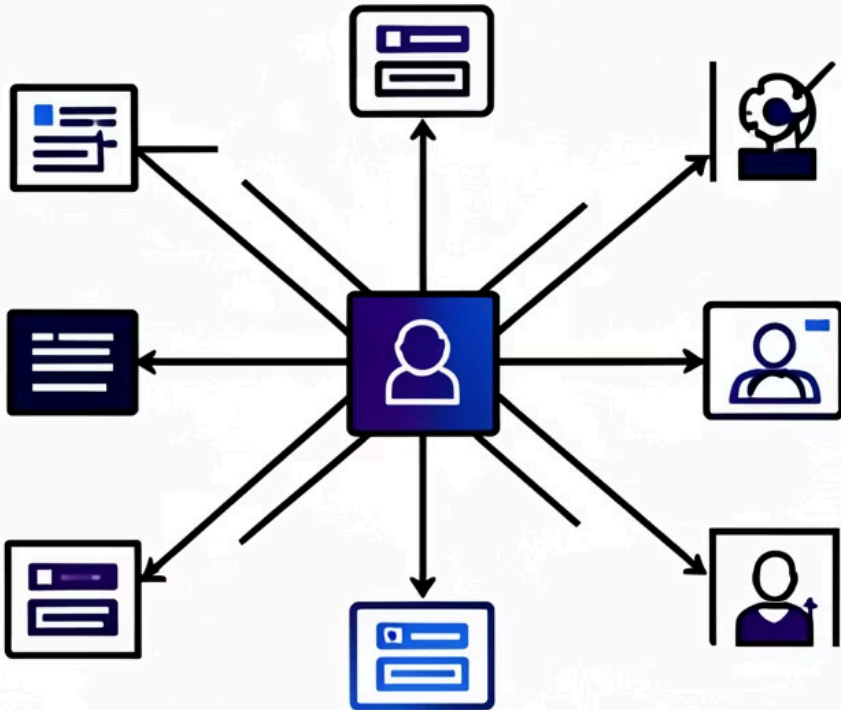
Deep Learning Model

Predicts the numerical salary value based on structured features.

04

Result

Presents the prediction to the user.



AI Agent Implementation

Model Used: Gemini Flash / Pro 2.5



Coordinating Agent

Orchestrates workflow and connects data from various sources.



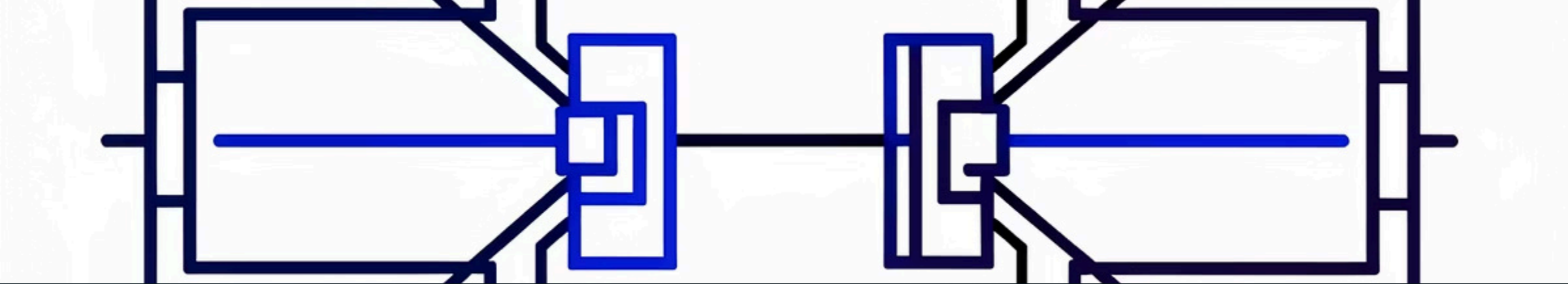
Data Extraction Agent

Parses job descriptions into a normalized format. Retrieves missing company information from a dataset/internet (using Google Search) to fill gaps in the reduced feature set.



Statistical Agent (planned)

Analyzes and explains statistical correlations between specific features (e.g., skills, location) and predicted salary. Provides the user with context *why* a particular salary was predicted.



Deep Learning Model Architecture

Dual Input Stream Design:

Input X1 (Numerical)

- Scaled continuous features
- Dense Layer (160 neurons, ReLU)
- Batch Normalization
- Dropout (0.4) for regularization

Input X2 (Categorical)

- Scaled binary/dummy variables

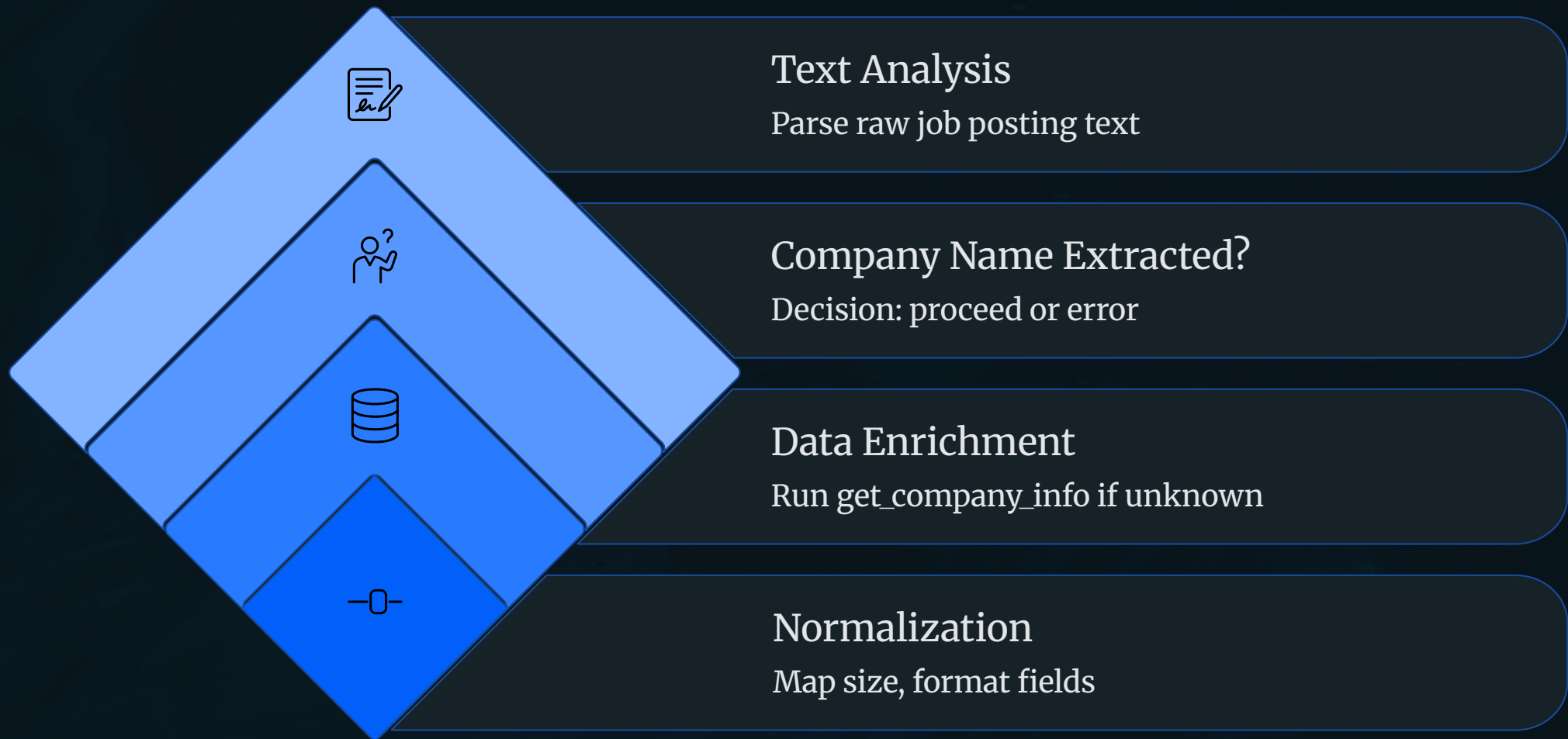
Fusion and Output

- X1 output passed to 2nd Hidden Layer (176 neurons, ReLU, Batch Norm, Dropout)
- Concatenation with X2
- **Output:** Dense(1, Sigmoid)

❏ **Training and Optimization:** Keras Tuner with Bayesian optimization determined layer sizes. Loss function: MSE. Optimizer: Adam.

Data Enrichment Pipeline

Our sophisticated pipeline processes raw job posting data, enriches it with external information, and normalizes it for the prediction model.



Quantitative Results

12K

Mean Absolute Error

MAE in USD – key
performance metric

15%

Percentage Error

Mean relative prediction error

29%

High Accuracy

Predictions with error < \$2K

66%

Good Accuracy

Predictions with error < \$10K

Results Interpretation: Considering the high variance in salary data and the subjective nature of job postings, we consider the average error of ~15% to be a strong result. Visual assessment (scatterplot) confirmed a strong linear correlation between predicted and actual values.

Leveraging AI in Project Development

We leveraged Generative AI tools to accelerate and enhance the development process:

Prompt Engineering and Optimization

AI helped refine instructions for agents, ensuring better data extraction and minimizing hallucinations.

Architectural Consultations

We received suggestions regarding neural network design (layer depth, number of neurons) and regularization techniques.

Presentation Creation

This presentation (structure and LaTeX code) was generated with the help of AI.

UI Development Support

AI assisted in generating boilerplate code and styling for the frontend user interface.

Future Work

- **Agent Improvement**
Refining Gemini Flash / Pro 2.5 prompts for even higher extraction accuracy.
- **Integration**
Tighter coupling between agents and the DL model for faster inference.
- **Expansion**
Expanding the dataset to other job sectors beyond Data Science.

Thank You

For your valuable time and interest in our work.

We believe this AI-driven approach offers a significant step forward in transparent salary insights.

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